

CLASS ASSIGNMENT

Section A - Short Answer Questions

Q1. What is meant by cognition? Name any two cognitive processes involved in problem solving.

Answer: Cognition means the mental processes we use to understand, learn, remember and use information. Two cognitive processes involved in problem solving are reasoning and memory.

Q2. List the four steps of Polya's problem-solving framework in the correct order.

Answer: The four steps are: (1) Understand the problem, (2) Devise a plan, (3) Carry out the plan, and (4) Look back / verify the answer.

Q3. What is functional fixedness? Give one suitable example.

Answer: Functional fixedness is when we think of an object only according to its normal use. For example, a person may see a coin only as money and not think of using it as a temporary screwdriver for a slotted screw.

Q4. Differentiate between functional fixedness and mental set in one or two points.

Answer: Functional fixedness limits the way we think an object can be used. A mental set makes us keep using a familiar method or strategy, even when another method may work better.

Q5. What is the basic idea behind Means-Ends Analysis (MEA)?

Answer: Means-Ends Analysis compares the current state with the desired goal. The difference is reduced by breaking the main problem into smaller subgoals and taking suitable actions to complete them.

Section B - Application-Based Questions

Q6. Applying Polya's Theory

A rectangular garden has a perimeter of 50 m and a length of 15 m. Find its width and area.

1. Understand the problem: We are given perimeter = 50 m and length = 15 m. We have to find the width first and then the area.

2. Devise a plan: Use the perimeter formula $P = 2(L + W)$ to find the width. After that, use $\text{Area} = L \times W$.

3. Carry out the plan:

$$50 = 2(15 + W)$$

$$25 = 15 + W$$

$$W = 10 \text{ m}$$

$$\text{Area} = 15 \times 10 = 150 \text{ m}^2.$$

4. Look back / verify: Perimeter = $2(15 + 10) = 2 \times 25 = 50 \text{ m}$. This matches the given perimeter, so the answer is correct.

Width = 10 m and Area = 150 m^2 .

Q7. Identify the Problem-Solving Barrier

a) Barrier: The cognitive barrier shown here is a mental set.

b) Why it occurs: The student is used to solving maths problems by directly applying formulas. Since this method worked before, the student keeps trying the same approach instead of thinking about another method such as drawing a diagram.

c) How to overcome it: The student should try to consider more than one method before solving. For example, they can ask whether a diagram, table, pattern or simpler example would make the problem easier to understand.

Section C - Analytical / Higher-Order Question

Q8. Means-Ends Analysis

The desired goal is to develop a simple student attendance management program that calculates the attendance percentage of each student and identifies students whose attendance is below 75%.

Current state: We have the students' attendance data, but their attendance percentages and shortage status have not been calculated yet.

Desired goal: Calculate the attendance percentage of every student and identify all students whose attendance is below 75%.

Subgoal 1 - Collect the required data: Take the student name/ID, number of classes attended and total number of classes held.

Subgoal 2 - Calculate attendance percentage: For each student, use:

Attendance % = (Classes Attended / Total Classes) x 100

Subgoal 3 - Compare with 75%: Use a condition to check whether the calculated attendance percentage is less than 75.

Subgoal 4 - Display the result: Show each student's attendance percentage and mark the students below 75% as Below 75%.

Means / actions from one state to the next:

Input attendance data -> Calculate percentage -> Compare with 75% -> Display result

Why dividing the problem helps: Breaking the main task into smaller subgoals makes the program easier to understand and develop. Each part can be solved and checked separately, and errors are easier to find.